

Solution: We know that the matrix equation $Ax=0$ is consistent since it has the trivial solution only. By Theorem 3 on Chapter 1, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.

Since the system of equations has a unique solution, By Theorem 2 from Chapter 1, there are no free variables. By defn, each of the n columns are pivot columns. By defn, a pivot position of a matrix corresponds to the leading 1 of a row in rref.

By the Row Reduction Algorithm and Theorem 1 from Chapter 1, the matrix A can be converted to a unique rref.

By Condition 4 of rref, each of the leading entries of a row must be 1. By Condition 2 of rref, for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it. Since there are n pivot columns and each row can have at most one leading entry, the leading entries of A must be on the main diagonal.

By Condition 3 of rref, for each pivot column, all positions below the pivot must contain 0. By Condition 5 of rref, for each pivot column, all positions above the pivot must contain 0 for each pivot column.

$\therefore$ The rref of A is I $\square$

24) Suppose A is $n \times n$ and the equation $Ax=b$ has a solution for each b in $\mathbb{R}^n$. Explain why A must be invertible.

Solution: By Theorem 4 from Chapter 1, Each row has a pivot. By defn, a pivot position of a matrix corresponds to the leading 1 of a row in rref. By Theorem 1 from Chapter 1 and the Row Reduction Algorithm, A can be converted to a unique rref. By Condition 4 of rref, each leading entry of a row must be 1. By Condition 2 of rref, for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it. $\therefore$ Each of the n columns are pivot columns since each of them contain exactly one pivot.